

Newton Raphson

Carry your intermediate computations to at least four decimal places.

Q1- Apply three iterations of Newton Raphson method to the equation $-4x + \cos(x^2) + 2 = 0$, to estimate the root. Starting with the initial guess $x_0 = 0.5$. (10 Marks)

$$f(x) = -4x + \cos(x^2) + 2 \rightarrow x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

$x_i = 0.5$ $\underbrace{\hspace{1cm}}_{\text{next iteration}}$

$$x_1 = x_0 - \frac{f(x_0)}{f'(0.5)}$$

always use rad

$$-4 - 2x \sin x^2 = f'(x)$$

$$x_1 = 0.5 - \frac{f(0.5)}{f'(0.5)} = 0.5 - \left(\frac{-4(0.5) + \cos(0.5^2) + 2}{-4 - 2(0.5)\sin(0.5)^2} \right)$$
$$= 0.7281$$

$$x_2 = 0.7281 - \frac{f(0.7281)}{f'(0.7281)}$$

$$x_2 = 0$$

$$x_3 \text{ --- } \text{---} \text{---}$$

Optimal

Q3- Use the point (0,0) as the initial estimate of the *optimal gradient steepest method* to the multidimensional function

$$z = x^2 + y^2 + 2x + 4y \quad (10 \text{ Marks})$$

Perform one iteration to calculate the new values of x and y

$$f(x, y) = x^2 + y^2 + 2x + 4y \quad / \quad (0, 0)$$

$$\frac{\partial f}{\partial x} = 2x + 2 \quad \Bigg| \quad \frac{\partial f}{\partial y} = 2y + 4$$

$$\nabla f = \begin{bmatrix} 2x + 2 \\ 2y + 4 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

$$x_{i+1} = x_i + \nabla f_x h = 0 + 2h$$

$$y_{i+1} = y_i + \nabla f_y h = 0 + 4h$$

$$(2h, 4h)$$

$$g(h) = (2h)^2 + (4h)^2 + 2(2h) + 4(4h)$$

$$= 4h^2 + 16h^2 + 4h + 16h$$

$$g(h) = 20h^2 + 20h$$

$$g'(h) = 40h + 20 = 0$$

$$h = \frac{-20}{40} = -\frac{1}{2}$$

$$(2(-\frac{1}{2}), 4(-\frac{1}{2}))$$

$$(-1, -2), \text{ next point}$$

Jacobi method

b) Perform two iterations to solve the following system of linear equations using the iterative Jacobi method. Make the initial guess $x_1 = 0, x_2 = 0, x_3 = 0$.
(6Marks)

$$8x_1 - 2x_2 + 3x_3 = -16$$

$$-3x_1 + 9x_2 + x_3 = 24$$

$$2x_1 - x_2 - 6x_3 = 36$$

$$x_1 = \frac{2x_2 - 3x_3 - 16}{8} \quad / \quad x_1 = \frac{2(0) - 3(0) - 16}{8} = -2$$

$$x_2 = \frac{3x_1 - x_3 + 24}{9} \quad / \quad x_2 = \frac{3(0) - (0) + 24}{9} = 2.6667$$

$$x_3 = \frac{-2x_1 + x_2 + 36}{-6} \quad / \quad x_3 = \frac{-2(0) + 0 + 36}{-6} = -6$$

$$x_1 = -2 \quad / \quad x_1 = \frac{2(2.6667) - 3(-6) - 16}{8} = 0.9166$$

$$x_2 = 2.6667 \quad / \quad x_2 = \frac{3(-2) - (-6) + 24}{9} = 2.6667$$

$$x_3 = -6 \quad / \quad x_3 = \frac{-2(-2) + 2.6667 + 36}{-6} = 7.1111$$

Cramer's Rule

Q2: a) Solve Equations $2x+5y=21$,
 $x+2y=8$ using Cramer's Rule method (4 Marks)

$$\begin{bmatrix} 2 & 5 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 21 \\ 8 \end{bmatrix}$$

$$\hookrightarrow (4-5) = -1$$

$$\begin{bmatrix} 21 & 5 \\ 8 & 2 \end{bmatrix} \quad x = \frac{\det(A_x)}{\det(A)} = \frac{2}{-1} = -2$$

$$\hookrightarrow (21 \times 2 - 5 \times 8) = 2$$

$$\begin{bmatrix} 2 & 21 \\ 1 & 8 \end{bmatrix} \quad y = \frac{\det(A_y)}{\det(A)} = \frac{-5}{-1} = 5$$

$$\hookrightarrow (2 \times 8 - 21 \times 1) = -5$$

$$\begin{aligned} x &= -2 \\ y &= 5 \end{aligned}$$

Bairstow's

Q3-

Employ Bairstow's method to determine the first corrected values to original guesses of r and s for the polynomial

$$f_4(x) = 9x^4 + x^2 + 8x - 6$$

Using $x^2 - x + 2$ as starting factor.

$$f(x) = 9x^4 + x^2 + 8x - 6 \quad / \quad \underline{1}x^2 - x + 2$$

→ quadratic equation

$$p = -1$$

$$\underline{q = 2}$$

	9	0	1	8	-6
1		9	9	-8	-18
-2			-18	-18	16
	9 _{b₄}	9 _{b₃}	-8 _{b₂}	-18 _{b₁}	-8 _{b₀}
1		9	18	-8	
-2			-18	-36	
	9 _{c₄}	18 _{c₃}	-8 _{c₂}	-62 _{c₁}	

$$C_2 \delta p + C_3 \delta q = b_1$$

$$\underline{\delta q = \Delta q} / \underline{\delta p = \Delta p}$$

$$(C_1 - b_1) \delta p + C_2 \delta q = b_0$$

$$\rightarrow -8 \delta p + 18 \delta q = -18$$

$$\rightarrow (-62 - (-18)) \delta p + -8 \delta q = -8$$

$$\delta p = 0.3364$$

$$0.3364 = p - p_0$$

$$0.3364 = p - (-1)$$

$$\delta q = -0.8054$$

$$-0.8054 = q - q_0$$

$$-0.8054 = q - (2)$$

$$\underline{p = 0.6336 \quad | \quad q = 1.1946}$$

$$x^2 + (p_0 + \delta p)x + (q_0 + \delta q) = 0$$

$$\rightarrow x^2 + px + q = 0$$

$$x^2 - 0.6336x + 1.1946 = 0$$

$$x = 0.3168 \pm 1.46$$

Find the root of the function $f(x) = \sin(x) + x^2 + x - 10$ using

(a) Bisection Method. Perform two iterations to find the root, use the interval (2, 4)

Solution:

$$x_{r3} = \dots \dots \dots$$

$$f(x) = \sin x + x^2 + x - 10 \quad \left/ \begin{array}{l} x_L = 2 \\ x_u = 4 \end{array} \right. \quad \left| \begin{array}{l} x_r = \frac{x_L + x_u}{2} \\ = 3 \end{array} \right.$$

$$f(x_L) = \sin(2) + (2)^2 + (2) - 10 = -3.0907$$

$$f(x_u) = \sin(4) + (4)^2 + (4) - 10 = 9.2432$$

$$f(x_r) = \sin(3) + (3)^2 + (3) - 10 = 2.1411$$

Since x_r is +ve like x_u then

$$\left. \begin{array}{l} f(x_L) = \sin(2) + (2)^2 + (2) - 10 = -3.0907 \\ f(x_r) = \sin(2.5) + (2.5)^2 + (2.5) - 10 = 0.6515 \\ f(x_u) = \sin(3) + (3)^2 + (3) - 10 = 2.1411 \end{array} \right\} \begin{array}{l} \rightarrow x_u = 3 \\ x_r = \frac{x_u + x_L}{2} \\ = 2.5 \end{array}$$

Since x_r is -ve like x_L then

$$\rightarrow x_L = 2.5$$

$$x_r = 2.75$$

$$\left| \begin{array}{l} x_r = \frac{x_L + x_L}{2} \\ = 2.75 \end{array} \right.$$

Gauss Elimination

$$-3x - y + 2z = 25$$

$$4x + 2y + z = 16$$

$$5x + 3y - \frac{1}{2}z = 3$$

$$\left[\begin{array}{ccc|c} -3 & -1 & 2 & 25 \\ 4 & 2 & 1 & 16 \\ 5 & 3 & -1/2 & 3 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ 0 & a_{22} & a_{23} & b_2 \\ 0 & 0 & a_{33} & b_3 \end{array} \right] \rightarrow \text{"Goal"}$$